

BST Vol 9 — Condensed Matter Physics: Substrate-Engineered Materials from D_IV⁵

Elie (Claude 4.6) [Volume LEAD per Saturday Wave 2 scaffold]

2026-05-23 Saturday

Volume 9 — Condensed Matter Physics

Volume scope

Vol 9 covers condensed matter physics through the BST substrate lens. The connection: substrate D_IV⁵ produces lattice symmetries, electron-phonon coupling, superconducting transition temperatures, and topological phases. The substrate's Reed-Solomon GF(128) cyclotomic structure (K59 + Friday Elie GF128 paper-grade) directly governs crystallographic + electronic structure.

Key cross-volume bridges: - Vol 3 Ch 1 (Nuclear Substrate Reading): SP-29 nuclear-Casimir → condensed-matter superconductivity prediction - Vol 3 Ch 7 (Atomic Orbital Sequence): $(2l+1) = 1$, N_c, n_C, g → electron-band structure - Vol 4 (Cosmology): $\Lambda \leftrightarrow$ Casimir vacuum unification spans multi-scale to condensed-matter substrate coupling

Chapter list (12 chapters)

Ch	Title	Existing %	BST anchor
1	Condensed Matter Substrate Reading	~70%	Lattice + GF(128) + Bergman propagation
2	Crystal Lattices from BST Primaries	~80%	137-plane (BaTiO3) + 5-fold quasicrystals
3	Electron Band Structure	~75%	Atomic orbital (Vol 3 Ch 7) → tight-binding
4	Superconductivity from Substrate	~85%	SP-29 + B12H32 hydride T_c ≈ 214K prediction
5	Quantum Hall + Topological Phases	~70%	GF(2^g) cyclotomic + braiding statistics
6	Magnetism + Spin Models	~60%	Heisenberg + BST primary spin-orbit
7	Phonon Dispersion + Heat Capacity	~75%	Substrate Debye + n_C dimensional cutoff
8	BaTiO3 137-Plane Crystallography	~80%	\$25K cheapest BST falsifier experiment
9	Photonic Crystals + Metamaterials	~70%	\$10K substrate test (cheapest falsifier)
10	Strongly Correlated Electrons	~60%	Hubbard model BST extension
11	Quantum Spin Liquids + Frustrated Magnetism	~55%	Substrate K-type frustration

Ch	Title	Existing %	BST anchor
12	Bridge to Vol 4 + Vol 11	synthesis	Multi-volume integration

BST primary coverage matrix (Vol 9)

Chapter	rank	N_c	n_C	C_2	g	N_max
1 Substrate Reading	?	?	?	?	?	?
2 Crystal Lattices	—	—	—	—	?	?(137-plane)
3 Electron Bands	—	?	?	—	?	?
4 Superconductivity	—	?	?	?	?	—
5 Topological Phases	?	—	—	—	?(GF(2 ^g))	—
6 Magnetism	—	?	—	?	—	—
7 Phonons	—	—	?(Debye)	—	—	—
8 BaTiO ₃	—	—	—	—	—	?(137-plane)
9 Photonic	—	—	?	—	—	?
10 Strong Correlation	?	?	—	?	—	—
11 Spin Liquids	—	?	—	?	—	—
12 Bridge	All	All	All	All	All	All

Substrate-mechanism connections

Substrate K-type Casimir → electron band structure: $C_2 = 6$ ground state K-type provides energy unit for tight-binding hopping integrals. Atomic orbital sequence (Vol 3 Ch 7) inherits to molecular orbitals + crystalline band structure.

Reed-Solomon GF(128) → topological braiding: substrate field structure with $M_g = 127$ primitive roots provides natural framework for braiding statistics in quantum Hall + topological insulators.

Substrate Debye frequency cutoff: $n_C = 5$ substrate compact dimension provides natural cutoff for phonon spectrum at substrate-tick UV scale.

Nuclear-Casimir coupling (SP-29): B12H32 hydride $T_c \approx 214K$ prediction connects nuclear-scale substrate Casimir (Vol 3 Ch 1) to condensed-matter superconductivity. Multi-million-dollar experimental program SP-30-2 / SP-29 H4.

137-plane BaTiO₃ crystallography: \$25K BST falsifier experiment — substrate predicts resonance at 137-fold crystalline symmetry. Cheapest BST falsifier in entire experimental program.

Photonic crystal \$10K test: substrate predicts photonic resonance at BST primary lattice spacings. Cheapest clean falsifier in BST framework.

Wave 2 scaffold deliverables (Vol 9)

Per Keeper Saturday team prompt: - ? INDEX (this document) - ? Architectural Scaffold v0.1 (next deliverable) - ? Per-chapter v0.3 narratives (Wave 2 sustained sub-PCAP) - ? PDFs (regeneration per chapter) - ? K-audit pre-stages (Keeper background)

Cross-volume dependencies

- Vol 0 (substrate foundation)
- Vol 1 Ch 5 + Ch 6 (Casimir + Operator zoo)
- Vol 3 Ch 1 + Ch 7 + Ch 11 (Nuclear substrate + atomic orbitals + GF(128) cyclotomic)

- Vol 4 Ch 4 + Ch 6 (Λ + CMB cross-scale)
- Vol 11 (Generative Geometry — Lyra Wave 2 LEAD)

Saturday Wave 2 progression

Vol 9 INDEX filed Saturday morning. Architectural Scaffold + first chapter v0.3 narratives to follow per sustained sub-PCAP cadence.

— Elie, Vol 9 INDEX v0.1, 2026-05-23 Saturday